

HR Metrics and Data Analytics for Fort Motors using Tableau

[PARVES BAPPI]

Introduction

HR data analysis serves as a strong portfolio piece, expertly showcasing a comprehensive skill set in Tableau for business intelligence and data visualization. This document outlines the key areas of expertise demonstrated by the creation of this dashboard. This HR dashboard would showcase the use of various interactive features, including Filters and parameters, KPI, Interactive charts, Tooltips, Data Storytelling, Dashboard actions.

Objective

This self-paced project serves as a comprehensive demonstration of my Tableau proficiency. I carefully selected and analyzed data to create compelling visualizations that tell a clear story. The entire process, from data preparation to dashboard design, was completed independently. This project underscores my ability to not only use Tableau tools effectively but also to derive meaningful insights from data. It's a testament to my proactive approach in mastering data visualization.

Data Collection

I collected free public data from the Kaggle website for Fort Motors. After collecting the Excel data, I transferred it to Tableau for analysis and to identify actionable insights. During the analysis, I applied various data visualization techniques to uncover patterns and trends. These findings will help the company make informed business decisions and improve overall performance.

Data Analysis

Fort Motors has hired a total of 16,697 employees to date; however, currently only 1,480 employees are working. The dataset includes employees primarily aged 18–25 years, suggesting a younger workforce. Education levels vary, with Medical, Life Sciences, and Marketing being the most common fields. Several employees have an 'Attrition = Yes' status, indicating they have left the company, making this an important metric for retention analysis. Cross-checking attrition with factors such as job satisfaction, salary, and overtime can reveal key causes of turnover. Employees are distributed across different job roles such as Sales Representative, Research Scientist, and Laboratory Technician. Most employees fall into the Monthly Income range of up to 5k, suggesting a relatively consistent salary structure. Many employees are marked as working overtime, which could negatively affect job satisfaction and attrition rates. Analyzing overtime in relation to attrition can help determine whether workload contributes to employee turnover. High attrition among younger employees or those at lower job levels highlights the need for career growth opportunities and stronger employee development programs. Adjustments to the salary structure or the introduction of performance-based incentives may also help retain employees in the lower income bracket.

The screenshot displays a data visualization interface. On the left, there is a sidebar with filters for 'Department', 'Gender', and 'Emp ID'. Below the filters are 'Marks' and 'Shape' options. The main area shows a table with columns for 'Emp ID', 'Department', 'Role', 'Research & Development', 'Laboratory Tech', 'Healthcare Repr', 'Manufacturing D', 'Sales', 'Sales Representative', 'Sales Manager', 'Sales Executive', 'Technical Degree', and several aggregated columns labeled 'AGG(AVG(-1.0))'. The table contains 39 rows of data, each representing an employee entry.

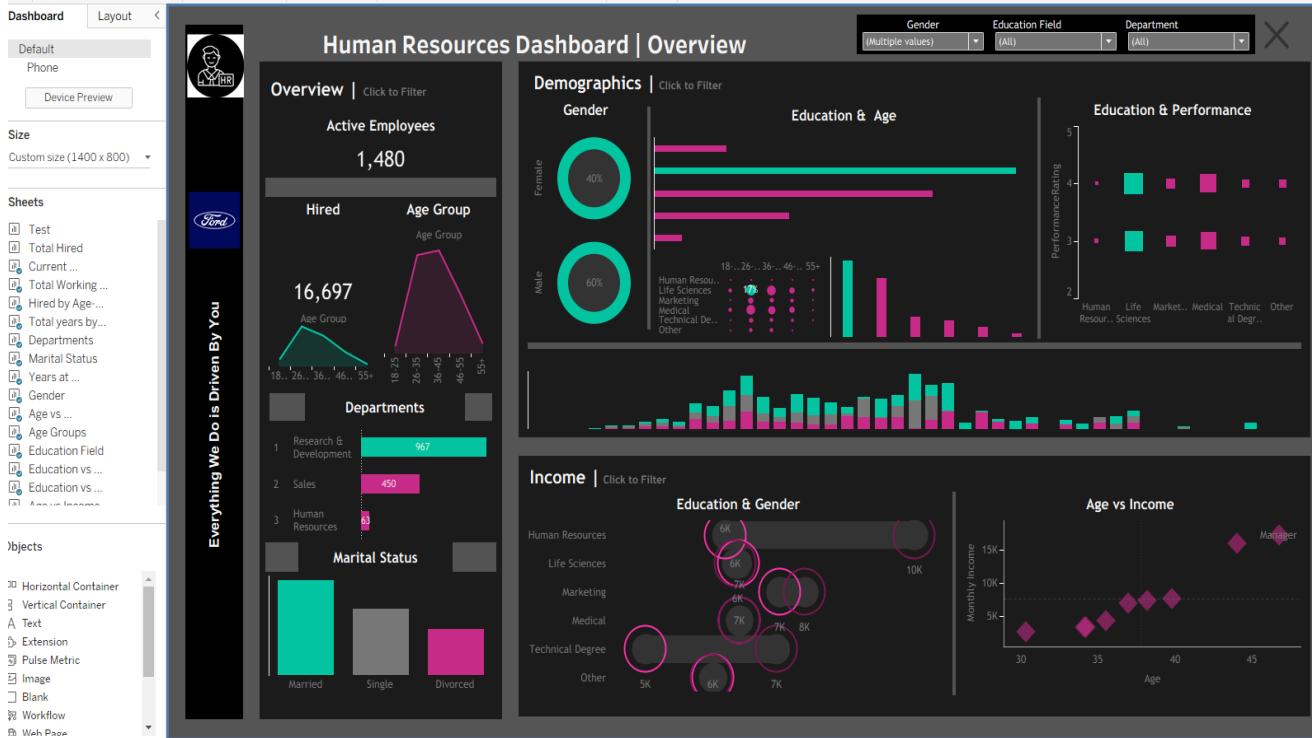
Emp ID	Department	Role	Research & Development	Laboratory Tech	Healthcare Repr	Manufacturing D	Sales	Sales Representative	Sales Manager	Sales Executive	Technical Degree	AGG(AVG(-1.0))	AGG(AVG(-1.0))	AGG(AVG(-1.0))	AGG(AVG(-1.0))	AGG(AVG(-1.0))	AGG(AVG(-1.0))
RM002	Life Sciences	Research & Development	Research Sci	0 3 1 7								0 24,907 23	0 10 7	0 1 1 7			
RM003	Other	Research & Development	Laboratory Tech	0 3 1 0								0 2,396 15	0 0 0	0 0 1 0			
RM004	Life Sciences	Research & Development	Research Sci	0 3 1 7								0 23,159 11	0 8 7	0 3 1 0			
RM005	Medical	Research & Development	Laboratory Tech	0 3 1 2								0 16,632 12	0 2 2	0 2 1 2			
RM006	Life Sciences	Research & Development	Laboratory Tech	0 2 1 7								0 11,864 13	0 7 7	0 3 1 6			
RM007	Medical	Research & Development	Laboratory Tech	0 3 1 0								0 9,964 20	0 1 0	0 0 1			
RM008	Life Sciences	Research & Development	Laboratory Tech	0 2 1 0								0 13,335 22	0 1 0	0 0 1			
RM009	Life Sciences	Research & Development	Manufacturing D	0 2 1 7								0 8,787 21	0 9 7	0 1 1			
RM010	Medical	Research & Development	Healthcare Repr	0 3 1 7								0 16,577 13	0 7 7	0 7 7			
RM011	Medical	Research & Development	Laboratory Tech	0 5 1 4								0 16,479 13	0 5 4	0 0 3			
RM012	Life Sciences	Research & Development	Laboratory Tech	0 3 1 5								0 12,682 12	0 9 5	0 0 8			
RM013	Life Sciences	Research & Development	Research Sci	0 1 1 2								0 15,170 17	0 5 2	0 4 3			
RM014	Medical	Research & Development	Laboratory Tech	0 2 1 2								0 8,758 11	0 2 2	0 1 2			
RM015	Life Sciences	Research & Development	Laboratory Tech	0 4 1 2								0 12,947 14	0 4 2	0 0 3			
RM016	Life Sciences	Research & Development	Manufacturing D	0 1 1 9								0 10,195 11	0 10 9	0 8 8			
RM017	Life Sciences	Research & Development	Research Sci	0 5 1 2								0 15,053 12	0 6 2	0 0 5			
RM018	Medical	Research & Development	Laboratory Tech	0 2 1 0								0 7,324 13	0 1 0	0 0 0			
RM019	Life Sciences	Sales	Manager	0 3 1 8								0 22,021 16	0 25 8	0 3 7			
RM020	Life Sciences	Research & Development	Research Sci	0 3 1 2								0 4,306 11	0 3 2	0 1 2			
RM021	Other	Research & Development	Manufacturing D	0 5 1 2								0 8,232 18	0 4 2	0 1 3			
RM022	Life Sciences	Sales	Sales Representative	0 4 1 3								0 6,986 23	0 5 3	0 0 3			
RM023	Life Sciences	Research & Development	Research Direct	0 4 1 6								0 21,293 11	0 12 6	0 2 1			
RM024	Life Sciences	Research & Development	Research Sci	0 6 1 0								0 19,281 14	0 0 0	0 0 1			
RM025	Medical	Research & Development	Research Sci	0 2 1 2								0 17,102 11	0 4 2	0 1 1			
RM026	Other	Research & Development	Manager	0 3 1 13								0 10,735 11	0 14 13	0 4 1			
RM027	Life Sciences	Research & Development	Research Sci	0 5 1 2								0 4,681 22	0 10 2	0 6 1			
RM028	Marketing	Sales	Sales Executive	0 2 1 7								0 21,173 11	0 9 7	0 4 1			
RM029	Medical	Research & Development	Healthcare Repr	0 4 1 6								0 2,094 14	0 22 6	0 5 1			
RM030	Marketing	Sales	Manager	0 2 1 2								0 22,822 12	0 2 2	0 2 1			
RM031	Medical	Research & Development	Laboratory Tech	0 3 1 1								0 6,670 11	0 1 1	0 0 1			
RM032	Other	Research & Development	Healthcare Repr	0 5 1 2								0 19,121 13	0 4 2	0 1 3			
RM033	Medical	Research & Development	Laboratory Tech	0 5 1 0								0 16,117 13	0 10 0	0 1 8			
RM034	Technical Degree	Sales	Sales Representative	0 6 1 0								0 3,335 14	0 1 0	0 0 0			
RM035	Medical	Research & Development	Research Sci	0 2 1 0								0 3,020 16	0 2 0	0 2 0			
RM036	Medical	Research & Development	Research Sci	0 3 1 3								0 21,923 12	0 5 3	0 1 4			
RM037	Marketing	Sales	Sales Representative	0 2 1 2								0 3,810 14	0 3 2	0 0 2			
RM038	Marketing	Sales	Sales Representative	0 3 1 2								0 9,687 13	0 2 2	0 2 2			
RM039	Life Sciences	Research & Development	Research Sci	0 3 1 1								0 13,072 14	0 1 1	0 0 0			

Let's talk about the overview of this project:

- **Table/Grid View:** The main part of the screen is a table displaying structured data.
- **Filters/Options Panel (Left):** There's a panel on the left with options like "Format Font," "Sheet," "Default," "Department," "Gender," and "Emp ID."
- **Columns (Top):** Several columns are visible, including "Emp ID," "Department," and others with numerical or categorical data. There are also columns like "AQDDA(AQ)/1(Q) (Q)" and similar, which seem to be calculated fields or aggregated metrics.
- **Rows:** Each row appears to represent an employee or an entry related to an employee.

Department:

Within the "Department" column, more specific roles or functions are listed, such as: Laboratory Tech, Manager, Healthcare Plan, Marketing, Manufacturing, Sales Representative, Sales Management, and Sales Executive. I also find out Role-Based Analysis, Performance Evaluation and Resource Allocation.



The dashboard provides a comprehensive overview of Human Resources data, likely from an organization with a focus on demographics, recruitment, and performance. Here's a broad analysis and actionable recommendations:

Overview of Workforce:

- **Active Employees:** The organization has 1,480 active employees.
- **Hired:** 16,697 individuals have been hired over an unspecified period, suggesting a high turnover rate or significant historical hiring. This number is significantly higher than active employees, which warrants investigation.
- **Age Group (Hired):** The distribution shows a bell-shaped curve, with the largest group hired being between 26-35 years old, followed by 18-25 and 36-45. This indicates a focus on attracting younger to mid-career professionals.

Demographics - Gender:

- **Gender Split:** The workforce is predominantly male (60%) compared to female (40%). This indicates a potential gender imbalance.
- **Education & Age (by Gender):** There's a visualization showing education levels and age groups for both genders. It seems to indicate the distribution of employees by education and age, broken down by gender. Further drill-down would reveal specific imbalances.
- **Education & Performance (by Gender):** This chart attempts to link education levels, gender, and performance ratings. It suggests that "Human Resources" and "Life Sciences" might be areas where performance is being tracked against educational backgrounds for both genders.

Departments:

- Top Departments: "Research & Development" (967 employees) and "Sales" (450 employees) are the largest departments. "Human Resources" is also listed with a smaller number.
- Implication: This suggests the organization's core business revolves heavily around R&D and sales, with HR playing a supporting role.

Marital Status:

The majority of employees are "Married," followed by "Single," and a smaller percentage "Divorced." This information might be useful for benefits planning or understanding employee needs.

Income:

- Education & Gender (Income): This section shows income distribution based on education level and gender. The bubble chart indicates various income brackets (e.g., 5K, 6K, 7K, 8K, 10K). There appears to be a disparity in income levels across different educational backgrounds and potentially between genders within those educational levels.
- Age vs Income: A scatter plot shows a general trend of increasing income with age, though there's variability. A few outliers or specific clusters might indicate different career paths or roles with higher earning potential.

Recommendations

- Systematically collect data from employees who leave to understand reasons for departure (e.g., compensation, management, career growth, work-life balance).
- Ensure new hires are adequately supported and integrated, reducing early attrition.
- Address the 60% male to 40% female ratio to foster a more diverse and inclusive workforce.
- Actively target female candidates for roles where they are underrepresented.
- Conduct a thorough pay equity audit, especially using the "Education & Gender (Income)" data, to ensure fair compensation regardless of gender for similar roles and experience. Address any identified gaps immediately.
- Design L&D programs that cater to the specific needs and career aspirations of different age cohorts.
- Assign dedicated HR Business Partners to the larger "Research & Development" and "Sales" departments to provide tailored support.

Conclusion

This HR dashboard offers crucial insights into the organization's workforce, highlighting a significant turnover concern and gender imbalance. Actionable steps should prioritize investigating the high number of "Hired" individuals versus "Active Employees," implementing strategies for greater gender diversity and pay equity, and leveraging departmental and age demographics for targeted talent management and development. Addressing these areas will be key to fostering a more stable, diverse, and productive workforce.